Dr. SIDRA Aslam

Email	sidraaslam742@gmail.com
ORCID	0000-0001-7020-1762
Google Scholar	https://scholar.google.com/citations?user=j84IgvcAAAAJ&hl=en

Professional Summary

I am an accomplished Computer Science researcher and former **Postdoctoral Fellow** at Shenzhen University, China. I also hold a **PhD with Highest Distinction** from the University of Primorska, Europe. My expertise encompasses high-priority cybersecurity domains, including data privacy and blockchain, supported by a strong record of publications in Q1-ranked journals such as the IEEE Internet of Things Journal **(Impact Factor 8.9).** With extensive experience in higher education and contributions to major European projects like H2020, I specialize in communicating complex decentralized security solutions to international audiences. I am dedicated to advancing research-led teaching and contributing to institutional strategic growth.

Education

10-2023 – 09-2025
Shenzhen, China

Postdoctoral Fellow

Shenzhen University

Research: I addressed the issues on data privacy in Internet of Things (IoT) healthcare. I designed and implemented a novel machine learning-based solution that achieved the key results. I published my research in IEEE IoT Journal (a high impact factor, Q1 ranked journal).

06-2019 – 06-2022
Koper, Slovenia

PhD in Computer Science | Passed with Highest Distinction

University of Primorska

- **Major:** Information Security and Privacy, Blockchain, Restful web services, Semantic Web, Web of Things, Distributed architecture
- **Thesis:** Decentralized Data Privacy and Security for the Web of Things
- Awarded Slovenian Government Merit Scholarship for Ph.D. Studies (2019-2022)

04-2018 – 06-2019
Saarbrücken, Germany

Promotion Masters in Computer Science

Graduate School of Computer Science | SAARLAND University

- **Major:** Data privacy & Semantic Web Technologies

09-2014 – 01-2017
Islamabad, Pakistan

Masters in Computer Science | Passed with Highest Distinction

COMSATS University

- **Major:** Information Security and Privacy, Access Control, Semantic Web Technologies, and Cloud Computing
- **Thesis:** Agent-based Identification and Classification of ROP (Role, Object, Permission) in Distributed Environment" address the problem of user authentication using agent and RBAC model

09-2009 – 09-2013
Wah cantt, Pakistan

BS (Hons) in Computer Science

University of Wah

Thesis: Macro Recording System

Aug 2007 – April 2009
Fateh Jang, Pakistan

Intermediate in Computer Science
Community Model Girls Higher Secondary School, Fateh Jang, Pakistan
Major: Physics, Mathematics and Computer Science

April 2005 – Mar 2007
Fateh Jang, Pakistan

Matriculation in Science
Govt. Girls High School, Fateh Jang, Pakistan
Major: Physics, Chemistry and Biology

Experience

09-2019 – 05-2022
Izola, Slovenia

Assistant Researcher
InnoRenew CoE company
Contribution:
- I worked on industrial projects (such as ARRS J2-2504, COST Action CA19126, and PHARAON H2020), conducted scientific research work, solved research problems, implemented results of scientific research work, cooperated in scientific activities and research work of the institute, and published high-quality research papers in Journals and conferences

10-2020 – 03-2021
Koper, Slovenia

Teaching Assistant
University of Primorska
Contribution:
- I prepared content to deliver high quality lectures to master's level students' module "data engineering and distributed information systems". I supervised students' semester projects and marked assessments.

06-2021 – 07-2021
Budapest, Hungary

Short Term Scientific Mission (STSM) Research work
Advanced Building and Urban Design (ABUD) company
Contribution:
- I conducted scientific research work, solved scientific research problems, implemented solutions and published research papers in Journals and conferences.

Research Projects

09-2019 – 05-2022
Izola, Slovenia

ARRS (National Slovenian Research Agency) project J2-2504 Autonomic Edge Computing for Air Quality Monitoring | InnoRenew CoE company

06-2020 – 05-2022
Izola, Slovenia

PHARAON H2020 (European funding) project number 857188 | InnoRenew CoE company

Memberships of Scientific Societies, Reviews

10-2024 – 10-2024
Reviewer of Journal of Scientific Reports

02-2020 – 11-2021
Management Committee Substitute of COST Action CA19126 Positive Energy Districts European Network.

12-2016 – 12-2016
Islamabad, Pakistan
Organizing Committee Member of Frontiers of Information Technology (FIT), 2016 14th International Conference

Talks Given

23-06-2025 Shenzhen, China	"Guarding Health Data: A New Machine Learning-Blockchain Framework for IoT" Invited speaker |College of Electronics and Information Engineering, Shenzhen University
14-03-2024 Shenzhen, China	"Blockchain Technology and Applications" Invited speaker | College of Electronics and Information Engineering, Shenzhen University
26-10-2021 Slovenia (online during covid)	"Privacy Preserving Decentralized Energy Management" Invited Speaker | COST Action CA19126 Positive Energy Districts European Network
24-06-2021 Budapest, Hungary	"Blockchain Mutability and Multi-level Access Control" Visited Speaker | Advanced Building and Urban Design (ABUD) company,
06-05-2021 Izola, Slovenia	"A Decentralized Framework for Data Updates and Privacy Management" Invited Speaker|InnoRenew CoE company
12-04-2021 Koper, Slovenia	"Blockchain Security, Privacy Issues and Challenges" Invited Speaker | Information and Computer Technologies (ICT) lab, UP Famnit
24-10-2018 Saarbrucken, Germany	"Data Privacy in Social Networks" |CISPA, Saarland University

Publications

2025	**Sidra Aslam**, Saba Aslam "A Hybrid On-Chain/Off-Chain Architecture for Secure Agri-Food Traceability: Integrating IPFS for Scalable Data Storage" ***In Progress Journal Article***
2025	**Sidra Aslam** et.al "A novel and secure machine learning-based hyperledger blockchain for IoT healthcare" IEEE Internet of Things Journal (Q1), (IF 8.9), 2025.
2023	**Sidra Aslam**, Michael Mrissa "A framework for privacy-aware and secure decentralized data storage" Computer Science and Information Systems Journal, (IF 1.8), 2023
2022	Michael Mrissa, Aleksandar Tošić, Niki Hrovatin, **Sidra Aslam**, Balázs Dávid, Lazslo Hajdu, Miklós Krész, Andrej Brodnik, Branko Kavsek "Privacy-aware and Secure Decentralized Air Quality Monitoring" SCI Journal Applied Sciences (Q1), (IF 2.838), 2022
2021	**Sidra Aslam**, Viktor Bukovszki, Michael Mrissa "Decentralized Data Management Privacy-aware Framework for Positive Energy Districts" SCI Energies MDPI Journal (Q1), (IF 3.004), 2021
2021	**Sidra Aslam**, Aleksandar Tosic, Michael Mrissa "Secure and Privacy-Aware Blockchain Design: Requirements, Challenges and Solutions" Journal of Cybersecurity and privacy, (IF 3.275), 2021
2020	**Sidra Aslam**, et.al "OBAC: Towards Agent-based Identification and Classification of ROP (Roles, Objects, Permissions) in Distributed Environment" SCI Journal of Multimedia Tools and Applications, (IF 2.313), 2020
2017	**Sidra Aslam**, et.al "Information Collection Centric Techniques for Cloud Resource Management: Taxonomy, Analysis, and Challenges" SCI Journal of Network and Computer Applications (Q1), (IF 6.281), 2017

Conferences

2021 Oslo, Norway	**Sidra Aslam**, Viktor Bukovszki, Michael Mrissa "Multi-Level Data Access Control in Positive Energy Districts" Sustainability in energy and buildings (SEB 21) conference on "Positive Energy District" 2021
2021 Tartu, Estonia	**Sidra Aslam**, Michael Mrissa "A RESTful Privacy-Aware and Mutable Decentralized Ledger" MADEISD workshop in the European Conference on Advances in Databases and Information Systems, 193-204, 2021
2021 Izola, Slovenia	**Sidra Aslam,** Michael Mrissa "Mutable and Privacy-aware Decentralized Ledger for Data Management in Wood Supply Chain Environments" InnoRenew CoE International Conference (IRIC2021), 41-41, 2021
2020 Portorož, Slovenia	**Sidra Aslam**, Michael Mrissa "Privacy-aware Distributed Ledger for Product Traceability in Supply Chain Environments" Conference of SWST International Society of Wood Science and Technology, 2020
2015 Rawalpindi, Pakistan	**Sidra Aslam**, Munam Ali Shah "Load Balancing Algorithms in Cloud Computing: A Survey of Modern Techniques" Software Engineering Conference National. IEEE, 2015

Conferences Presentations & Certifications

2021 Oslo, Norway	Presented research paper titled "Multi-level Data Access Control in Positive Energy Districts" in SEB 21 conference on "Positive Energy Districts".
2021 Tartu, Estonia	Presented research paper titled "A RESTful Privacy-aware and Mutable Decentralized Ledger" MADEISD workshop in the European Conference on Advances in Databases and Information Systems.
2021 Izola, Slovenia	Presented research paper titled "Mutable and Privacy-aware Decentralized Ledger for Data Management in Wood Supply Chain Environments", InnoRenew CoE International Conference (IRIC2021).
2020 Portorož, Slovenia	Presented research paper titled "Privacy-aware Distributed Ledger for Product Traceability in Supply Chain Environments", Conference of SWST International Society of Wood Science and Technology.
2018 Saarbrücken, Germany	Participated in the European Data Science Summer School at the Saarland University.
2016 Islamabad, Pakistan	Participated as an Organizing Committee Member of Frontiers of Information Technology (FIT), 14th International Conference held at Serena Hotel.
2016 Islamabad, Pakistan	Attended workshop on "Web Mining & Semantic" organized by Frontiers of Information Technology (FIT).
2015 Rawalpindi, Pakistan	I presented research paper titled "Load Balancing Algorithms in Cloud Computing: A Survey of Modern Techniques" in IEEE National Conference on Software Engineering (NSEC).

2013 Rawalpindi, Pakistan	I participated as a member of ICT COMPPEC project exhibition at NUST college of EME.

Achievements

06-2021 – 07-2021 Budapest, Hungary	Awarded grant for Short Term Scientific Mission (STSM) COST Action: CA19126
06-2019 – 06-2022 Koper, Slovenia	Awarded Slovenian Government Scholarship for study PhD Computer Science at the University of Primorska
04-2018 – 05-2019 Saarbrücken, Germany	Awarded German Government Scholarship for study MS Computer Science at Graduate School of Computer Science, Saarland University
09-2014 – 01-2017 Islamabad Pakistan	Received a laptop in the Prime Minister Laptop Scheme at COMSATS University
2015 Rawalpindi, Pakistan	Awarded Best Research Paper Shield at National Software Engineering Conference (NSEC), IEEE

Relevant Skills

- **Core Technical Expertise:** I am specialized in high-priority cybersecurity domains, including **Data Privacy**, **Blockchain Technology**, and **Information Security**.
- **Research & Analytical Tools:** She demonstrates proficiency in advanced research tools such as **LaTeX**, **Mendeley**, and **Protégé** for semantic web modeling.
- **Programming & Data Science:** She possesses strong programming skills in **Python**, **Java**, and **JavaScript**, complemented by data science libraries including **NumPy**, **SciPy**, and **Pandas**.
- **Web & Distributed Technologies:** Her expertise extends to **Semantic Web (OWL)**, **HTML/CSS**, and distributed architectures for the **Web of Things (WoT)**.
- **Project & Academic Leadership:** She has a proven ability to contribute to major international frameworks, including **H2020** and **ARRS** projects.
- **Communication & Mentorship:** Dr. Sidra is an experienced educator who has delivered master's level lectures and frequently presents complex security solutions as an **invited speaker** at international institutions